

Name:

Qousain Haider

Roll-No:

2024-CET-13

Department:

Software Engineering

Group:

Computer Engineering Technology

Semester:

4th Semester

Subject:

Database Applications

Lab Task:

SQL Server – DDL Commands

Submitted To:

Engr. Asma Javid

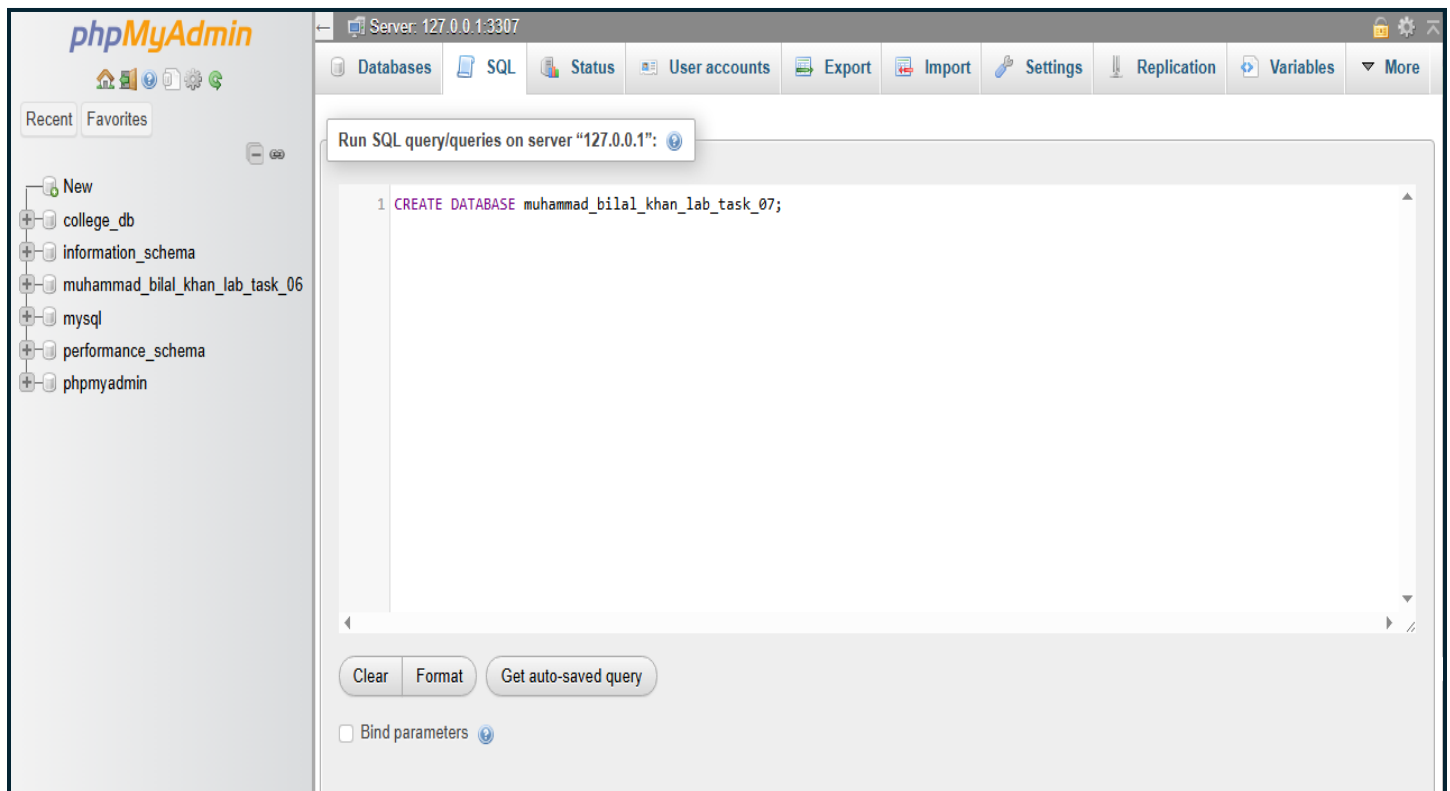
Lab Task 7

Section 1:

Creating a Database and Table Using SQL DDL Commands

1. Create a new database named as your full name in all lowercase with underscores, followed by _lab_task_07.

Example: kainat_pervaiz_lab_task_07





2. In this database, use SQL DDL commands to create a table named PUBLISHERS with the following attributes:

Attribute	Data Type	Field Size
P_ID	VARCHAR	4
P_Name	VARCHAR	30
Address	VARCHAR	40
State	VARCHAR	20
Phone	VARCHAR	20

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07

Structure SQL Search Query Export Import Operations Privileges Routines Events

Run SQL query/queries on database muhammad_bilal_khan_lab_task_07:

```
1 CREATE TABLE PUBLISHERS (  
2 P_ID VARCHAR(4),  
3 P_Name VARCHAR(30) NOT NULL,  
4 Address VARCHAR(40),  
5 State VARCHAR(20) NOT NULL,  
6 Phone VARCHAR(20)  
7 );
```

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - New
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: publishers

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0017 seconds.)

SELECT * FROM `publishers`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

P_ID	P_Name	Address	State	Phone
------	--------	---------	-------	-------

Query results operations

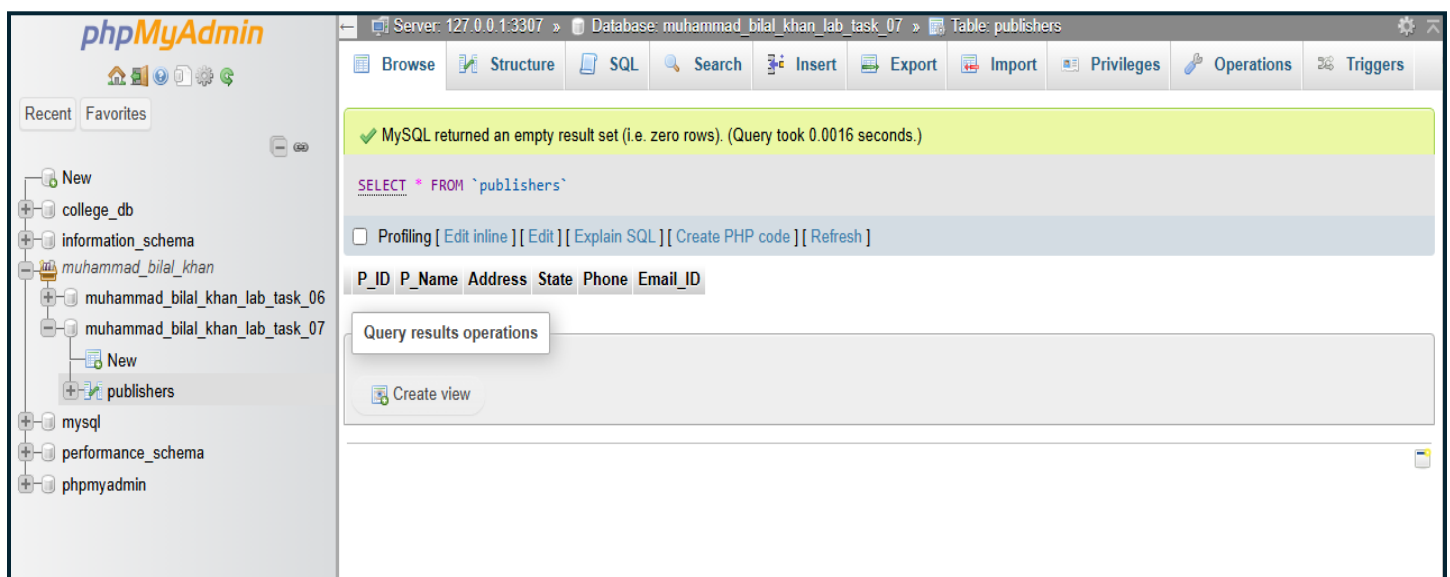
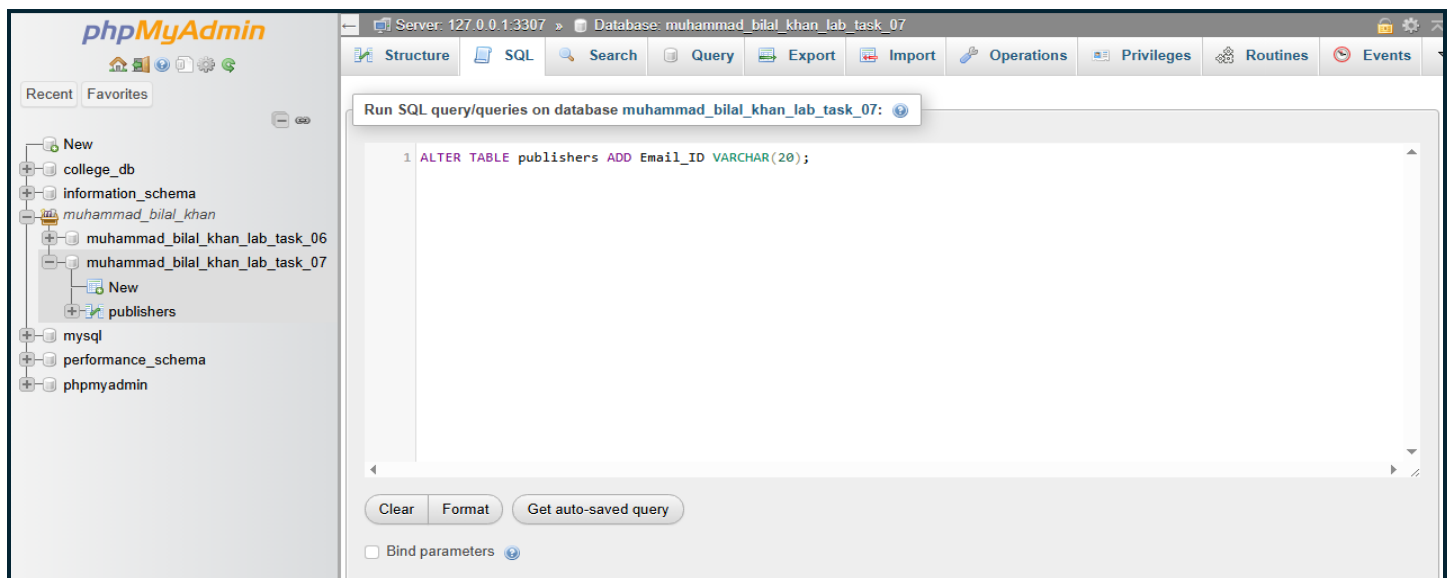
Create view

Section 02:

Modifying Table Structure Using SQL DDL Commands

Use SQL DDL commands to make the following changes to the PUBLISHERS table:

1. Add a new column Email_ID of type VARCHAR(20).



2. Change the data type of P_ID from VARCHAR to INT.

The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'muhammad_bilal_khan' and 'muhammad_bilal_khan_lab_task_07'. The 'publishers' table is selected. The main panel shows the 'SQL' tab with the following query:

```
1 ALTER TABLE publishers MODIFY P_ID INT;
```

On the right, the table structure is displayed:

P_ID	P_Name	Address	State	Phone	Email_ID
------	--------	---------	-------	-------	----------

At the bottom, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'.

The screenshot shows the phpMyAdmin interface with the 'Table structure' tab selected. The table structure is as follows:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	P_ID	int(11)			Yes	NULL			Change Drop More
2	P_Name	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
3	Address	varchar(40)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	State	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
5	Phone	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More
6	Email_ID	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Below the table structure, there are options for 'Check all', 'With selected', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', 'Spatial', and 'Fulltext'.

3. Change the data type of Phone from VARCHAR to INT.

The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'muhammad_bilal_khan' and 'muhammad_bilal_khan_lab_task_07'. The 'publishers' table is selected. The main panel shows the 'SQL' tab with the following query:

```
1 ALTER TABLE publishers MODIFY Phone INT;
```

On the right, the table structure is displayed:

P_ID	P_Name	Address	State	Phone	Email_ID
------	--------	---------	-------	-------	----------

At the bottom, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'.

phpMyAdmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: publishers

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	P_ID	int(11)			Yes	NULL			Change Drop More
2	P_Name	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
3	Address	varchar(40)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	State	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
5	Phone	int(11)			Yes	NULL			Change Drop More
6	Email_ID	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Check all With selected: Browse Change Drop Primary Unique Index Spatial

Fulltext

Print Propose table structure Move columns Normalize

Add 1 column(s) after Email_ID Go

Section 03:

Renaming and Deleting Table Using SQL DDL Commands

1. Rename the table PUBLISHERS to PUBLISHER using an SQL command.

phpMyAdmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: publishers

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table muhammad_bilal_khan_lab_task_07.publishers:

```
1 ALTER TABLE publishers RENAME TO publisher;
```

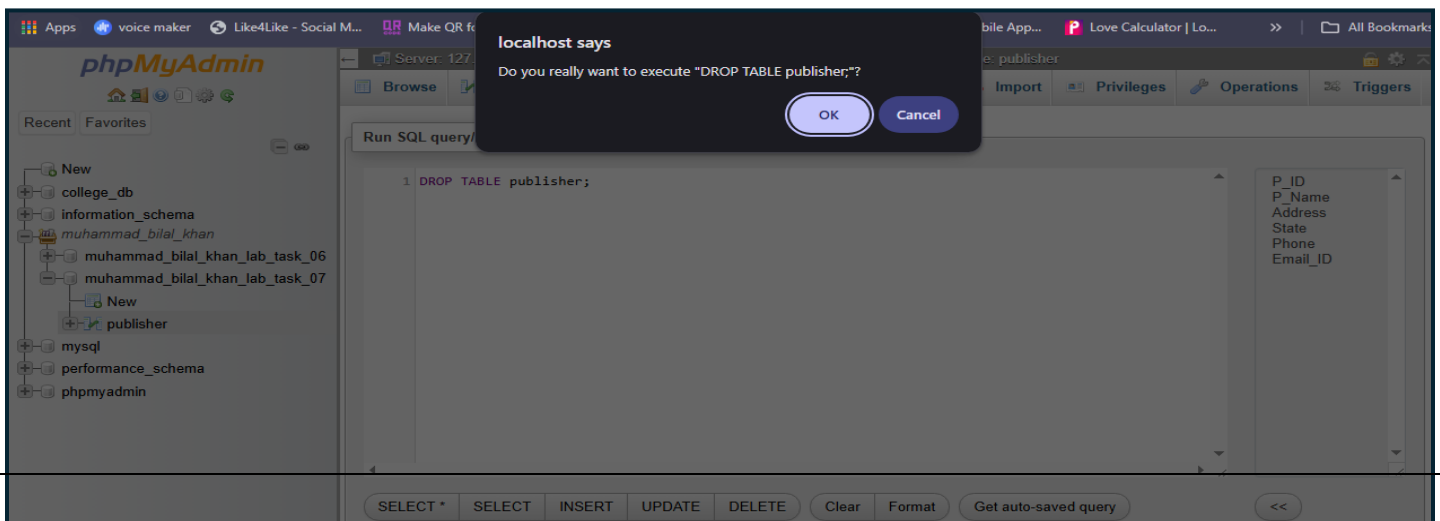
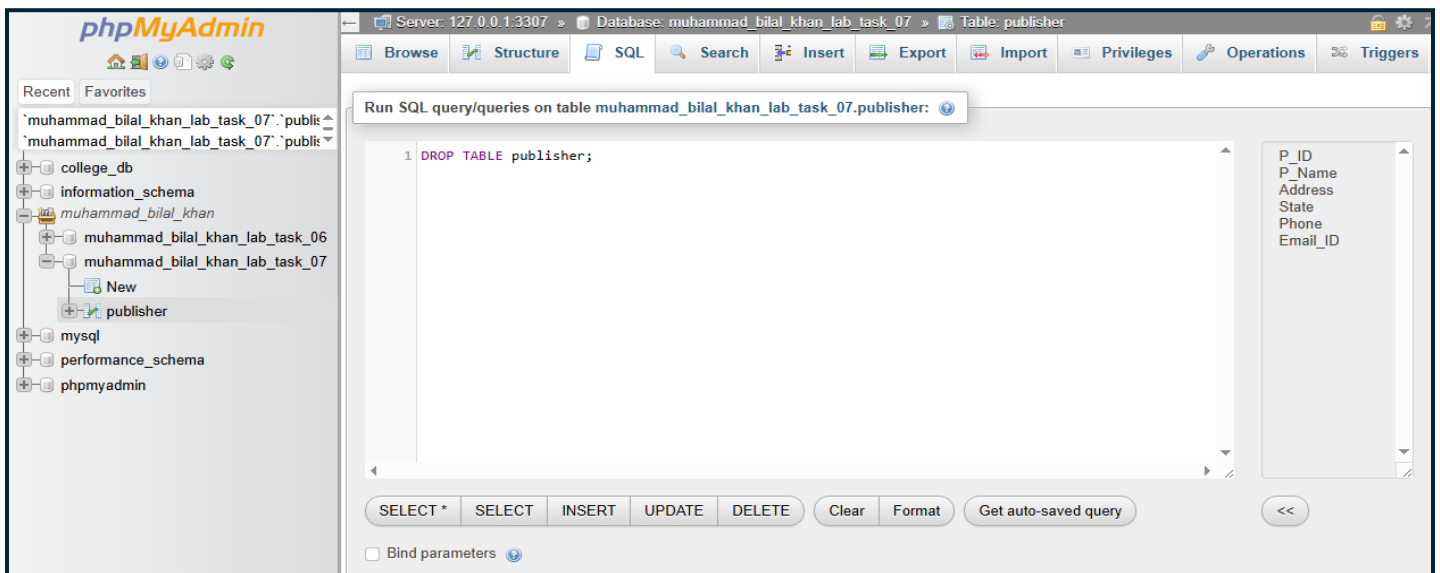
P_ID
P_Name
Address
State
Phone
Email_ID

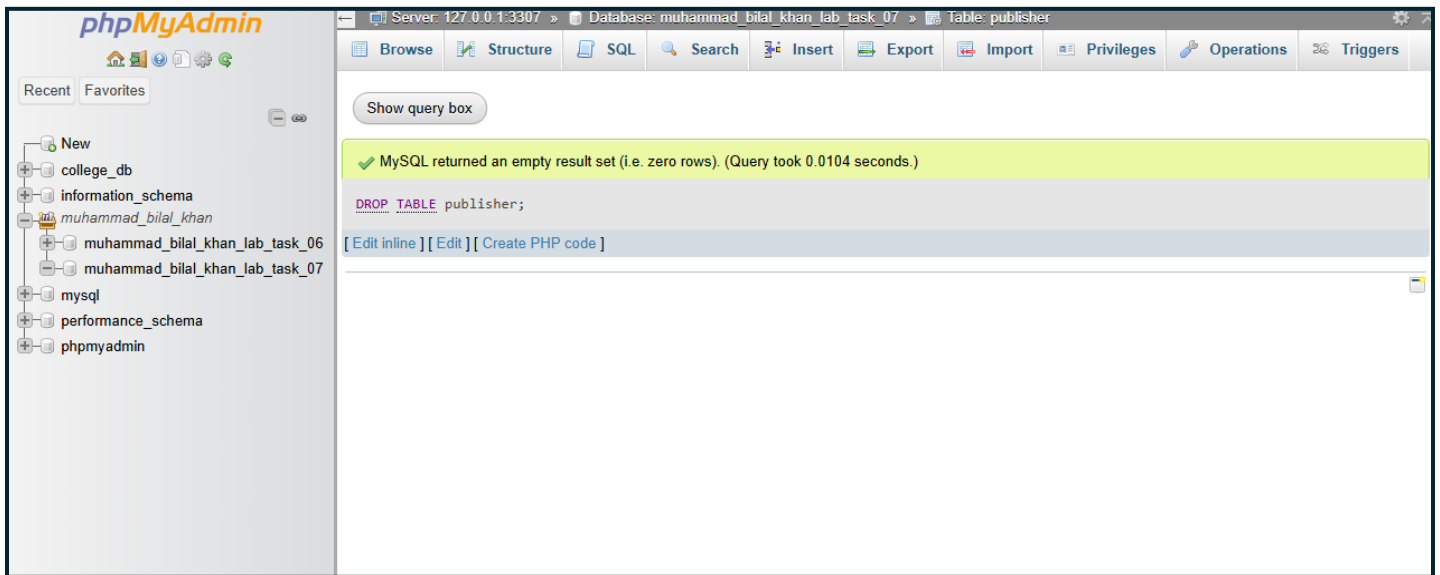
SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

Bind parameters



2. Delete the renamed table PUBLISHER using SQL.



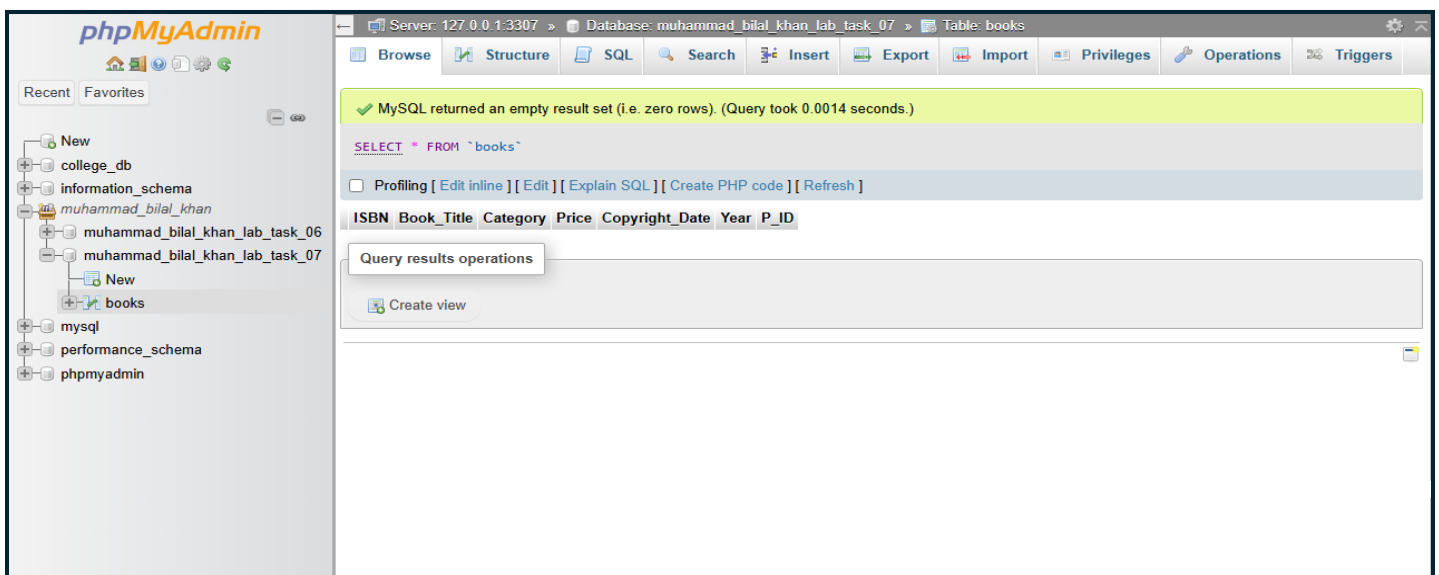
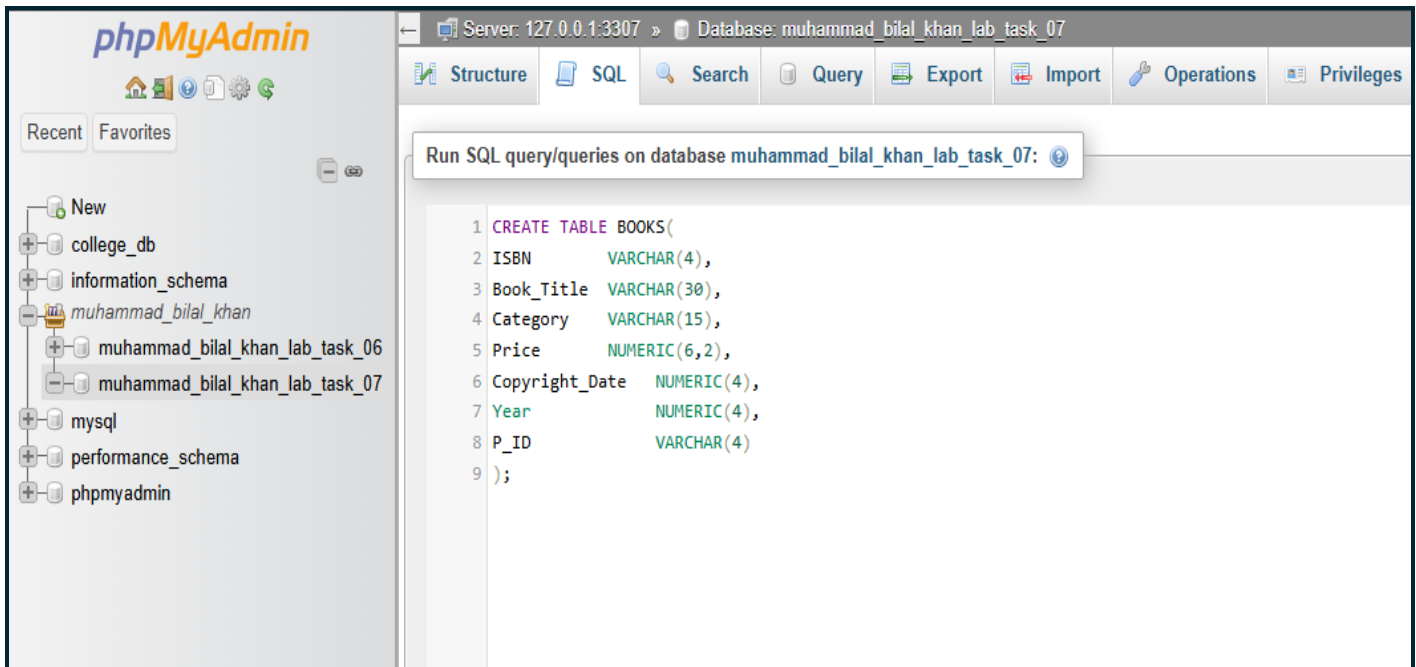


Section 04:

Creating and Modifying Another Table Using SQL DDL Commands

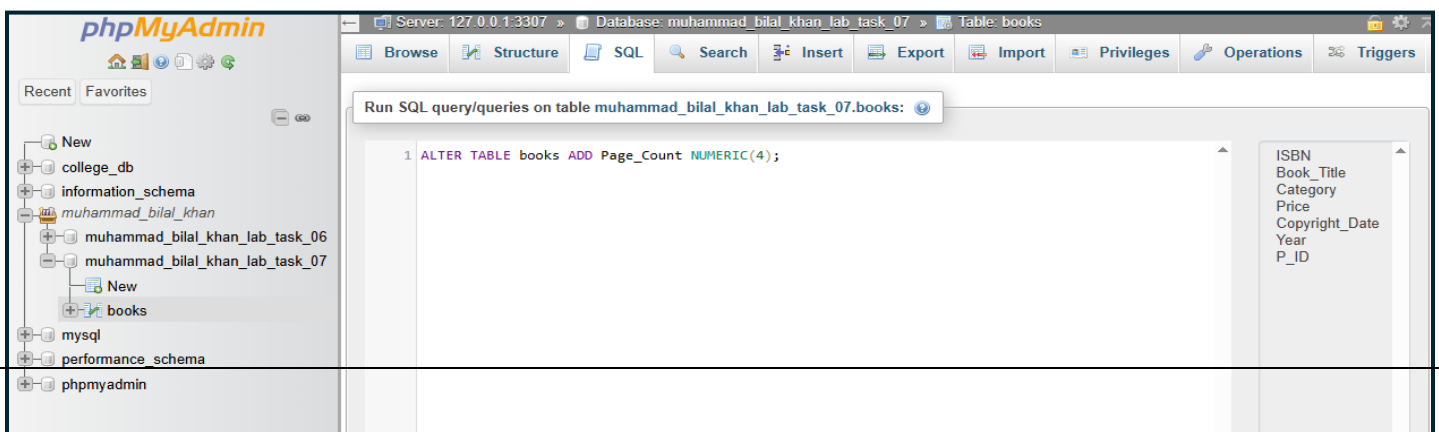
1.Create table BOOKS with the following structure:

Attribute	Data Type	Field Size
ISBN	VARCHAR	4
Book_Title	VARCHAR	30
Category	VARCHAR	15
Price	NUMERIC	(6,2)
Copyright_Date	NUMERIC	4
Year	NUMERIC	4
P_ID	VARCHAR	4



2.Modify the BOOKS table using SQL DDL Commands:

Add a new column Page_Count with type NUMERIC(4).



Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: books

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)

```
SELECT * FROM `books`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

ISBN	Book_Title	Category	Price	Copyright_Date	Year	P_ID	Page_Count
------	------------	----------	-------	----------------	------	------	------------

Query results operations

Create view

Change the data type of ISBN from VARCHAR to INT.

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: books

Run SQL query/queries on table muhammad_bilal_khan_lab_task_07.books:

```
1 ALTER TABLE books MODIFY ISBN INT;
2
```

ISBN
Book_Title
Category
Price
Copyright_Date
Year
P_ID
Page_Count

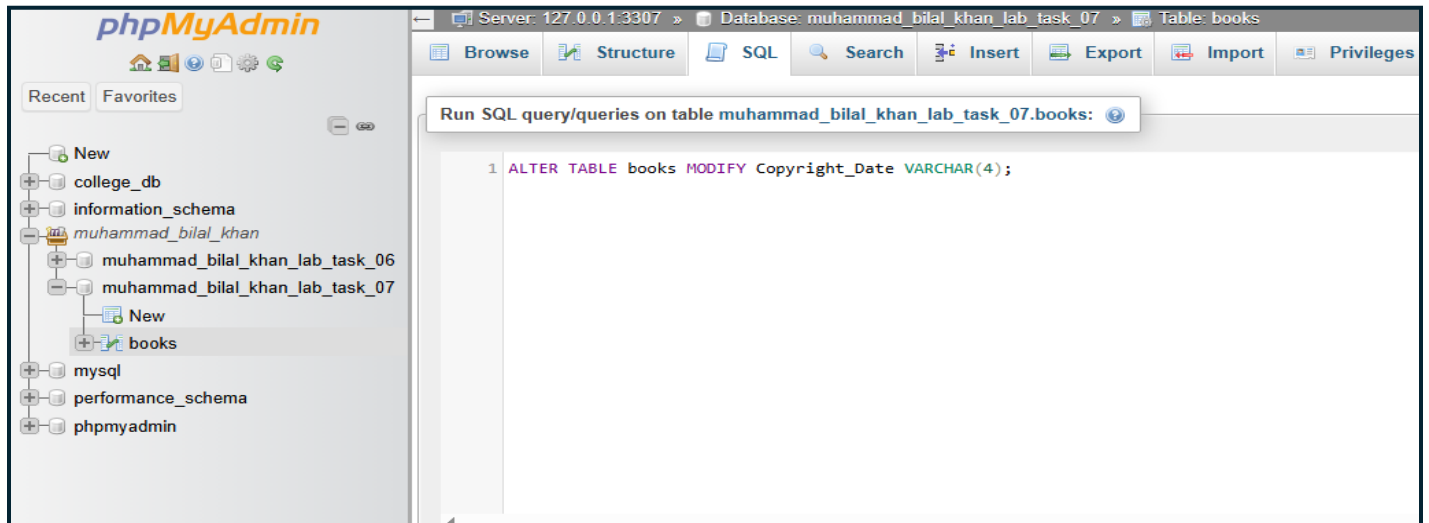
Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: books

Table structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 ISBN	int(11)			Yes	NULL			Change Drop More
☐	2 Book_Title	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	3 Category	varchar(15)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 Price	decimal(6,2)			Yes	NULL			Change Drop More
☐	5 Copyright_Date	decimal(4,0)			Yes	NULL			Change Drop More
☐	6 Year	decimal(4,0)			Yes	NULL			Change Drop More
☐	7 P_ID	varchar(4)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	8 Page_Count	decimal(4,0)			Yes	NULL			Change Drop More

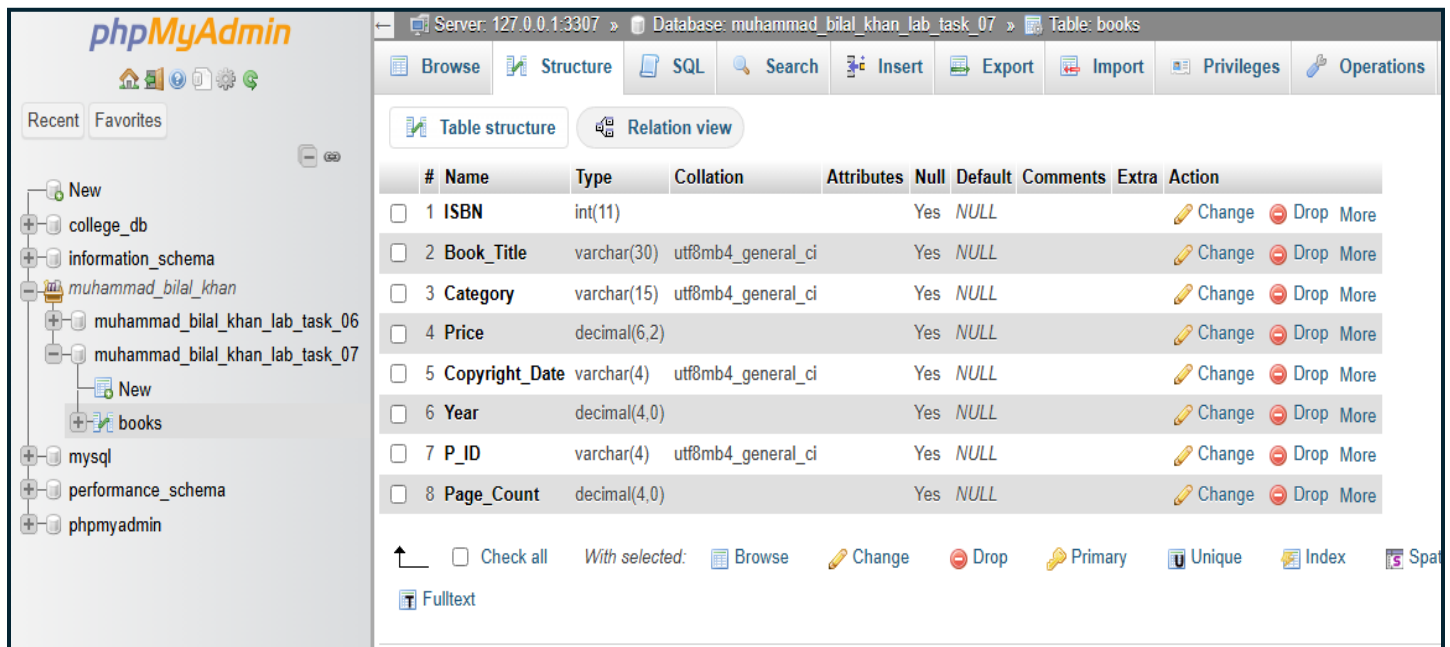
Check all With selected: Browse Change Drop Primary Unique Index Spatial

Change the data type of Copyright_Date from NUMERIC to VARCHAR.



The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'muhammad_bilal_khan_lab_task_07' and its table 'books'. The main panel is in the 'SQL' tab, showing a query to alter the 'books' table:

```
1 ALTER TABLE books MODIFY Copyright_Date VARCHAR(4);
```



The screenshot shows the phpMyAdmin interface with the 'Table structure' view selected for the 'books' table. The table structure is as follows:

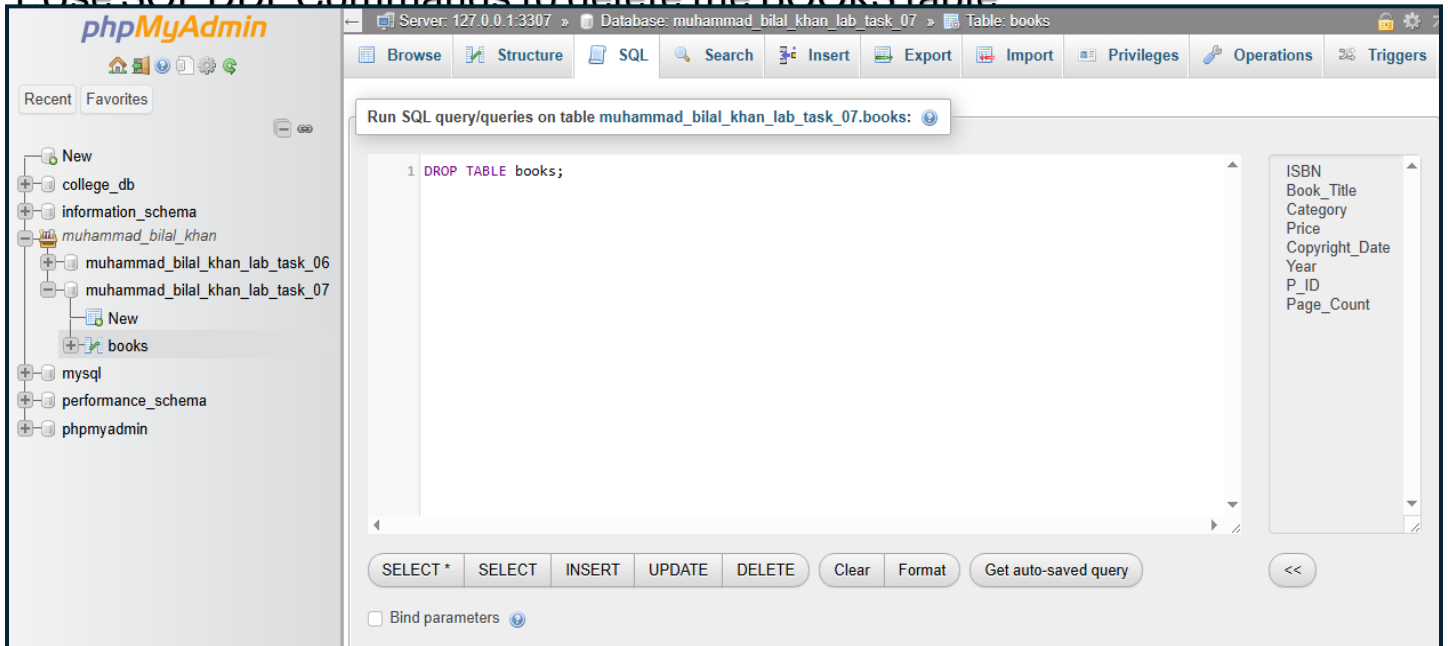
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	ISBN	int(11)			Yes	NULL			Change Drop More
2	Book_Title	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	Category	varchar(15)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	Price	decimal(6,2)			Yes	NULL			Change Drop More
5	Copyright_Date	varchar(4)	utf8mb4_general_ci		Yes	NULL			Change Drop More
6	Year	decimal(4,0)			Yes	NULL			Change Drop More
7	P_ID	varchar(4)	utf8mb4_general_ci		Yes	NULL			Change Drop More
8	Page_Count	decimal(4,0)			Yes	NULL			Change Drop More

At the bottom, there are options to 'Check all', 'With selected:', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', and 'Spatial'.

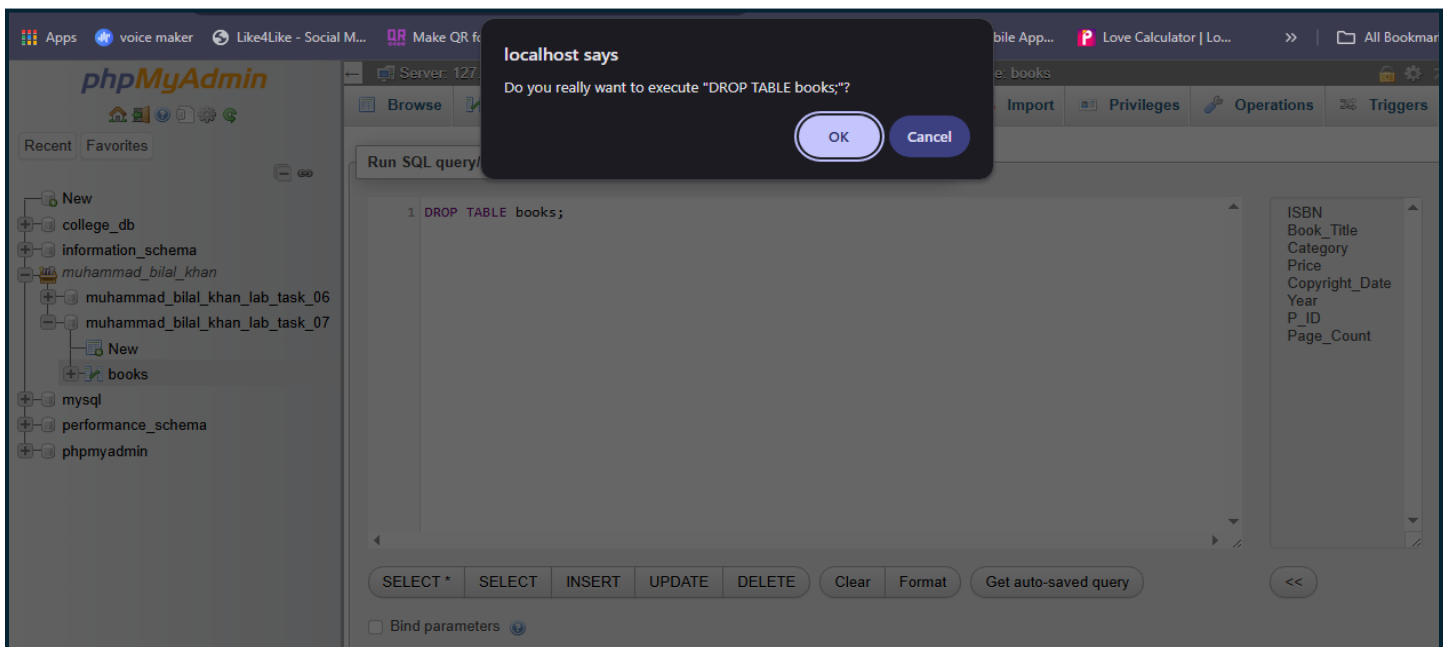
Section 05:

Deleting Tables and Database

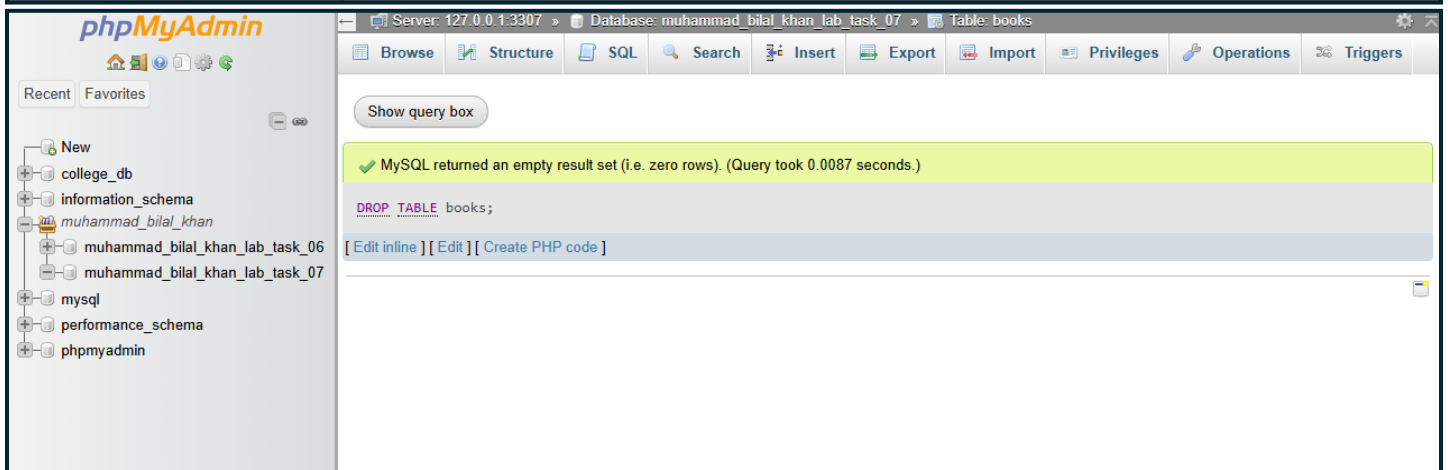
1 Use SQL DDL Commands to delete the BOOKS table



The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'college_db', 'information_schema', 'muhammad_bilal_khan', 'muhammad_bilal_khan_lab_task_06', 'muhammad_bilal_khan_lab_task_07', 'mysql', 'performance_schema', and 'phpmyadmin'. The 'books' table is selected under 'muhammad_bilal_khan_lab_task_07'. The main panel shows the SQL query editor with the command `DROP TABLE books;` entered. The table structure is listed on the right: ISBN, Book_Title, Category, Price, Copyright_Date, Year, P_ID, and Page_Count. Below the query editor are buttons for SELECT *, SELECT, INSERT, UPDATE, DELETE, Clear, Format, and Get auto-saved query. A checkbox for 'Bind parameters' is also present.



The screenshot shows the phpMyAdmin interface with a confirmation dialog box overlaid. The dialog box is titled 'localhost says' and contains the text 'Do you really want to execute "DROP TABLE books;?"'. There are 'OK' and 'Cancel' buttons. The background shows the same SQL query editor as the previous screenshot, with the command `DROP TABLE books;` entered.



The screenshot shows the phpMyAdmin interface after the successful execution of the `DROP TABLE books;` command. A green message box at the top states: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0087 seconds.)'. Below this, the command `DROP TABLE books;` is displayed, along with links for '[Edit inline]', '[Edit]', and '[Create PHP code]'. The table structure is no longer visible on the right side of the interface.

2. Delete the entire database from your system

(this can be done manually after taking a screenshot showing its deletion).

